

Project: Eco-Tracker

In efforts for certification of Eco-School at Tsinglan

Description:

Helps track food waste in school cafeterias and gamifies food waste, students earn eco-credits for not wasting food.

Features:

1. Food Waste Tracking Station

- Station in cafeteria, with a screen/camera(1) on wall directed horizontal, and camera(2) facing down directed down on the counter, atop counter is scale.
- 1. camera(1) - identity scan, scans user's face to identify them
- 2. scale - user places tray on top of scale to measure weight of food
- 3. camera(2) - (overhead) to capture image of tray compartments. user clicks on button to take photo
- **Scoring logic**
 - Tray: 5 compartments (4 small, 1 large – weighted 2× in scoring).
 - Each compartment scored:
 - 0 = no waste
 - 1 = a bit of waste
 - 2 = some waste
 - 3 = full waste
 - Final score = sum of compartments (large compartment counts twice).

2. Hardware + AI

- **Currently have a raspberri pi 5 and camera module 3 (which will serve as camera 2), need to find another camera w screen for the horizontal camera (1)**
- **need scale, that can communicate w raspberri pi**
- **AI vision model**
 - Object detection model trained to classify waste level per compartment.
 - Training via Label Studio (currently)
 - Images stored in Cloudflare R2 bucket for labeling, training, and audit logs.
- Real-time feedback: Screen displays score, eco-credits earned, and cumulative impact (e.g., "You saved 0.3kg of CO₂ today").

3. User Management

- Face-based authentication: Users register by scanning their face once, linking it to their account (username/English name). Future scans recognize them automatically.

4. UI Gamification & Rewards

- Eco-credits: Awarded to environmentally friendly actions, primarily not wasting food in this case, other ways to earn it can be featured later
- Leaderboards: all-time version and monthly, by individual, and class
- Achievements & badges – Milestones like “Zero Waste Week Streak” “First 1000 Credits.”
- **Reward redemption** – credits can be redeemed for prizes, which we can set up a small booth near the cafe, it can be stationary, school merchandise etc

5. Analytics & Reporting

- Dashboard – keep track of total waste trends